MILITARY SPECIFICATION

LINERS, FLYERS' ANTI-EXPOSURE COVERALL, CWU-72/P

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and five sizes of a continuous wear flyer's anti-exposure coverall liner designated as CWU-72/P.

1.2 Classification. The liners shall be furnished in the following sizes, as specified (see 6.2.1):

SIZES

Extra Small
Small
Medium
Large
Extra large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 9323, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

V-F-106	-	Fasteners, Slide, Interlocking.
UU-P-553	-	Paper, Wrapping, Tissue.
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams and Stitchings.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipping and Storage.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1580AS100	-	Index, Liner Assy., Flyer's Anti-Exposure Coverall, CWU-72/P.
1580AS101	-	Liner Assy, Flyer's, Anti-Exposure Coverall, CWU-72/P.
1580AS102	-	Innershell Subassy, Flyer's, Anti-Exposure Coverall Liner, CWU-72/P.
1580AS103	-	Outershell Subassy, Flyer's, Anti-Exposure Coverall Liner, CWU-72/P.
1580AS104	-	Belt Subassy, Flyer's Anti-Exposure Coverall Liner, CWU-72/P.
1580AS105	-	Label, Identification CWU-72/P.
1580AS191	-	Liner Patterns, Flyer's, Anti-Exposure Coverall, CWU-72/P, Size X-Small.
1580AS192	-	Liner Patterns, Flyer's, Anti-Exposure Coverall, CWU-72/P, Size Small.
1580AS193	-	Liner Patterns, Flyer's, Anti-Exposure Coverall, CWU-72/P, Size Medium.
1580AS194	-	Liner Patterns, Flyer's, Anti-Exposure Coverall, CWU-72/P, Size Large.
1580AS195	-	Liner Patterns, Flyer's, Anti-Exposure Coverall, CWU-72/P, Size X-Large.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the acquisition activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, samples of the CWU-72/P anti-exposure coverall, flyers' liners shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials and components. The materials and components shall conform to applicable specifications, standards and drawings as required herein. Unless otherwise specified, the materials and components shall have been manufactured not more than eighteen months prior to the date of delivery of the anti-exposure coveralls (see 6.2.2).

3.2.1 Slide fastener. The following requirements of V-F-106 for the slide fastener are also applicable (see 6.2.2):

a. Shrinkage control of the fastener tapes. (Ref. Paras. 3.3.11.1 and 4.4.19 of V-F-106)
b. Chain endurance. (Ref. Paras. 3.6.2 and 4.4.3.2 of V-F-106)

3.2.2 Tape. The end of the tape used to fabricate the slide fastener thongs shall be treated to prevent fraying. No sharp edges shall be formed.

3.3 Design. The liners shall be a one-piece garment with a front slide fastener opening conforming to Drawing 1580AS101, and shall be fabricated from an innershell and an outershell bound together at the edges. The innershell consists of a cloth covered fibrous thermal insulation. The liners are designed for wear under outer anti-exposure coveralls and therefore do not have any pockets.

3.4 Construction. The liners shall be constructed in accordance with Drawings 1580AS101, 1580AS102, 1580AS103 and 1580AS104.

3.4.1 Cutting. The liners shall be cut in strict accordance with the patterns, Drawings 1580AS191 through 1580AS195, as applicable, (see 6.4), which provide for seam allowance; directional lines for cutting; and notches or markings for the proper assembling of all the parts. The patterns provide a seam allowance of 3/8-inch. The Government patterns shall not be altered in any manner and shall be used for making the working patterns. The working patterns shall be identical to the Government furnished patterns. The component parts of the liner shall be cut from the same ply of the lay.

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3.4.2 Shade and size marking. The component parts of the liner shall be marked or ticketed by any commercial method, including bundling, to insure a uniform shade throughout the liner and for the proper assembly of the liner. No metal fastening device or sewn-on ticket shall be used. The shade and size markings may be accomplished by the use of an ink pad numbering machine or rubber stamp, provided the markings do not show through the outerside of the liners and are not deleterious to the cloth. The markings placed on the seam allowance shall not be visible on the outerside of the liners. The adhesive type shade and size marking tickets may be used for the shade and size markings (see 3.4.2.1). The assembled liner shall not contain any shade or size marking ticket nor any marking that was used for locating the position for attaching of any component.

3.4.2.1 Adhesive shade tickets. The adhesive shade tickets shall be fabricated from paper with a thermoactivated adhesive applied on one side. The adhesive shall not discolor the cloth and the adhesive mass shall not adhere to the cloth upon removal of the ticket. The heat used to attach the thermoactivated adhesive tickets shall not stiffen, harden, sear or damage the cloth in any manner.

3.4.3 Replacement of defective components and cut parts. Care shall be exercised during the spreading, cutting and manufacturing processes to assure that any component and cut part containing any defect or damage that is classified in this specification as a defect shall be replaced with a non-defective and properly matched component or cut part, as applicable.

3.4.4 Stitches, seams and stitching. Stitch Type 301 of FED-STD-751, with 8 to 10 stitches per inch, shall be used in the joining together of the component parts of the innershell (cloth covered thermal insulation), and of the belt subassembly and for the attachment of slide fastener, slide fastener reinforcing tape, fastener tapes, label, belt channel and the binding webbing to the liner. Safety Stitch Type 516 of FED-STD-751 with 9 to 12 stitches per inch, 3/8 inch wide, shall be used to join together the component parts of the outershell. The width of the overedge stitching portion of the safety stitch (Stitch Type 504) shall be 1/8 to 3/16 inch. The looper thread of the chain stitch portion of the safety stitch (Stitch Type 401) shall be on the innerside of the outershell. The innershell shall be formed by superimposing the thermal insulation components parts on the applicable cut parts of the innershell cloth and joining them together with Stitch Type 301 as specified in Drawing 1580AS102. The inner and outershells shall be joined together at the edges with overedge Stitch Type 504. The overedged sewn edges shall be sewn within the folded binding webbing so that the center or folded portion of the webbing is flush against the overedged sewn edges and the turned portions of the webbing are equidistant on each side of the liner edges. The binding webbing shall be sewn to the liner with Stitch Type 301. The seam allowances shall be maintained so that no runoff or open seam shall occur. All seams shall start and finish uniformly. No seam or component shall be twisted, puckered or pleated and no part of the liner shall be caught in an unrelated operation or seam. All the seam edges shall be properly forced out and shall not contain any folds. All the thread ends shall be trimmed to a maximum length of 1/4 inch. No cut or raw edge shall be visible on the interior or exterior of the liner. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching, in any row, shall be

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maintained within a tolerance of plus or minus 1/32 inch. The same tolerance for spacing and parallelism shall be applicable for the double row of stitches. The ends of the Stitch Type 301 stitching that are not caught in other seams or stitching shall be backstitched, by overlapping on themselves, a minimum of 1/2 inch. Each thread break, skip, and run-off shall be overstitched not less than 1 inch. Skipped stitches or thread breaks on on Stitch Types 401 and 504 stitching may be repaired using Stitch Type 301. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together.

3.4.4.1 Bartacking. The bartacking joining the belt subassembly to the outershell shall be 1 inch long centered on the elastic portion of the belt subassembly and shall contain a minimum of 56 zigzag stitches (Stitch type 304 of FED STD 751), 3/16 inch plus or minus 1/16 inch wide. A zigzag stitch is defined as the thread between two needle holes. Therefore, the number of zigzag stitches shall be determined by counting the number of needle holes and subtracting one.

3.5 Label. Each of the liners shall have a combination identification, size and washing instruction label conforming to Drawing 1580AS105. The label shall be located and attached to the liner as specified in Drawing 1580AS102.

3.6 Workmanship. The liners shall not contain any hole, abraded area, tear, cut, mend, patch, scorch or needle chew. The liners shall be uniform in quality and shall be free from irregularities or any defects that could adversely affect performance, reliability, or durability. Because of the emergency and life support use of the liners, the importance of providing a product of uniformly excellent quality cannot be overemphasized. The liners shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. The first article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.3 through 4.3.2 and 6.3).

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b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on the individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.4 through 4.4.1.3)

4.3 First article inspection. The first article inspection of the liners shall consist of examinations for all the requirements of this specification.

4.3.1 First article samples. Unless otherwise specified, as soon as practicable after award of the contract or order, the manufacturer shall submit three liners for each size of liner ordered in the contract, and a statement of certification (with supporting test data) that the slide fasteners conform to the chain endurance and shrinkage requirements cited in paragraphs 3.2.1a and 3.2.1b. The samples shall be representative of the construction, workmanship, components and materials to be used during production. When a contractor is in continuous production of these garments from contract to contract, submission of further first article inspection samples, on the new contract, may be waived at the discretion of the acquiring activity (see 6.2.1). Approval of the first article inspection samples or the waiving of the first article inspection does not waive the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1).

4.3.2 Disposition of first article samples. Upon completion of the first article inspection, recommendations and comments pertinent for use in monitoring production will be forwarded by the Government activity responsible for the inspection program (see 6.2.1) to the contracting officer. One approved sample of each size will be returned to the manufacturer for use in monitoring production. One sample will be consumed or destroyed during testing and will not be considered as part of the quantity to be delivered under the contract or order. The other sample will be retained by the test facility for reference.

4.4 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

Visual examination of the liners.
Examination at a laboratory.
Preparation for delivery.

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1.1 Liners. An inspection lot size shall be expressed in units of one liner made essentially under the same conditions and from the same materials and components. The sample unit shall be one liner.

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4.4.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing garments of one size, fully prepared for delivery from essentially the same materials and components as the remainder of the lot. The sample unit shall be one shipping container, containing garments of one size, fully prepared for delivery with the exception that it need not be sealed.

4.4.1.2 Sampling for visual examination of the liners and packaging. The sample size, acceptance criteria, and examinations required for the liners or the preparation for delivery, as applicable, shall be as specified in Table I, MIL-STD-105, and 4.4.1.3.

4.4.1.3 Sampling for examination of the liners at a laboratory. Upon completion of the examinations specified in Table I, a random sample shall be selected from each lot in accordance with MIL-STD-105, inspection level S-3. Each liner, selected as a sample unit, shall be forwarded to the laboratory specified in the procurement document (see 6.2.1) for a visual examination in accordance with the requirements and methods of Table I. The Government activity responsible for conducting the inspection program (see 6.2.1) shall report the results of tests and examinations to the designated inspection and acceptance office specified in the procurement document. Final acceptance of the lot from which the sample units were selected shall be based upon successful completion of the inspection program by the cognizant Government Quality Assurance Representative/Specialist, applying the applicable acceptance criteria specified in Table I. Any cost for retesting a lot, necessitated by the rejection of a previous lot, shall be borne by the manufacturer. Samples from a rejected lot shall not be resubmitted for examinations without the approval of the Contracting Officer. Upon completion of the examinations, the inspection facility shall return the garments to the manufacturer at the manufacturer's expense.

4.5 Inspection methods.

4.5.1 Visual examination.

4.5.1.1 Liners. Each liner, selected as a sample unit from the lot, shall be thoroughly checked dimensionally and examined visually to determine conformance to this specification. The classification and lists of defects, Table II, shall be used to classify and enumerate the defects found.

4.5.1.2 Packaging. Each of the shipping containers, selected as a sample unit from the lot, shall be examined visually to determine that the packaging, packing and marking conform to the requirements of this specification. The list of defects, Table III, shall be used to enumerate the defects found.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Industrial, as specified (see 6.2.1).

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5.1.1 Level A. Each liner with the slide fastener fully closed shall be neatly folded in accordance with good commercial practice and in such a manner to form a compact package without any damage to the slide fastener. Tissue conforming to UU-P-553, if desired, may be placed in the fold areas. The liner size on the label shall be clearly visible in front so as to permit ready identification. Each folded liner shall be individually inserted into a snug-fitting flat style, clear polyethylene film bag of 0.00125 inch thickness (plus/minus 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal as close as possible to the open end. A 1/4-inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.1.1 Intermediate packaging. Four liners of a single size only, packaged as specified in 5.1.1, shall be packaged in an intermediate snug-fitting container conforming to PPP-B-636, Style, Type and Variety optional, Class Weather-Resistant. Closure of the container shall be accomplished by adhesive bonding or tape-sealing (no staples permitted) in accordance with procedures specified in the appendix to PPP-B-636.

5.1.2 Industrial. The liners, folded as specified in 5.1.1, shall be preserved and packaged to afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity. The level of packaging may conform to the supplier's commercial practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be Level A, B or Industrial, as specified (see 6.2.1).

5.2.1 Level A. Sixteen liners of one size only, packaged as specified in 5.1.1.1, shall be packed in a wooden shipping container conforming to overseas type of PPP-B-601. The inside of each container shall be fitted with a taped liner conforming to Class Weather resistant type CF, grade V3 of PPP-F-320. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pounds minimum basis weight kraft paper. Each container shall be closed, waterproofed, and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification. If the quantity of liners of the same size is less than the quantity of liners required per container, mixed sizes (see 5.3.1) may be packed within the same shipping container.

5.2.2 Level B. Sixteen liners of one size only, packaged as specified in 5.1.1, shall be packed in a fiberboard shipping container conforming to Style RSC-L, Type CF or SF, Variety SW, Class Domestic, Grade 275 of PPP-B-636. The insid of each shipping container shall be fitted with a taped liner conforming to Type CF, Variety SW, Class Domestic, Grade 275 of PPP-F-320. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper. Each container shall be closed in accordance with appendix of the container specification. If the quantity of liners of the same size is less than the quantity of liners required per container, mixed sizes (see 5.3.1) may be packed within the same shipping container.

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5.2.3 Industrial. The liners shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special markings required by contract or order, interior and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Mixed sizes. Each shipping container that is packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "mixed sizes" plainly stamped or printed theron. Under these words shall be legibly stamped or printed the quantities and sizes of the liners therein.

6. NOTES

6.1 Intended use. The liners covered by this specification are intended for continuous wear by aircrewmembers under anti-exposure coveralls on overwater flights.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Government part number(s) and National Stock number(s), as applicable, and quantities.

c. Applicable drawings, including revisions.

d. First article requirements (see 4.3).

e. Name and address of the first article inspection laboratory (see 4.3.1); including the Government Activity responsible for conducting the inspection program (see 4.3.2).

f. Name and address of the production lot inspection laboratory (see 4.4.1.3) and the Government activity responsible for conducting the inspection program (see 4.4.1.3).

g. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

h. Whether any special markings are required (see 5.3).

i. Certificate of compliance for the age of materials and components (see 3.2 and 6.2.2).

j. Certificate of compliance of the slide fasteners to shrinkage control and chain endurance requirements (see 3.2.1 and 4.2.2).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirement	Applicable DID no.
3.2	Certificate of compliance for the age of materials and components	DI-E-2121
3.2.1	Certificate of compliance of the slide fastener to shrinkage control and chain endurance	DI-E-2121
4.3.1	First article inspection reports	DI-T-5329

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When first article inspection is required, the liners will be tested and should be a first preproduction item. The first article should consist of the samples specified in 4.3.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.4 Patterns. The patterns for the liners will be furnished by the contracting officer to the contractor for use in cutting working patterns (see 3.4.1). The quantity of cut parts shall be governed by the applicable pattern drawings.

Preparing activity:
Navy - AS
(Project 8475-N197)

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TABLE 1. Sample size, acceptance criteria, examinations, and tests of the liners and packaging.

Inspection	Paragraph		Sample size	Acceptance criteria
	Requirement	Method		
Visual examination	3.3 through 3.6 and Drawings 1580AS100 through 1580AS105 and Drawings 1580AS191 through 1580AS195, as applicable	4.5.1.1	Inspection Level II	An acceptable quality level of 2.5 defects per 100 units for major defects and 6.5 defects per 100 units for total (major plus minor) defects
Packaging	Section 5	4.5.1.2	Inspection Level S-2	An acceptable quality level of 2.5 defects per 100 units

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TABLE II. Classification of defects for the visual examination of the assembled liners.

Defect	Major	Minor
GENERAL		
a. Any hole, scissors or knife cut, tear, mend, patch, burn or weakening defect such as smash, multiple float, loose slub, needle chew, or abraded area	X	
b. Any portion of the liner stiffened, hardened, or scorched by any process of manufacturing	X	
c. Any binding short, tight or twisted causing puckers or twisting of any portion of the liner greater than 1/2 inch along any edge		X
d. Any missing, broken or coarse yarn, thin or thick place, misweave, woven in waste, loose or snagged yarn	1/	
SHADED PARTS		
a. Any part shaded or any shade bar		X
b. Color of any component not as specified or not uniform		X
CLEANNESS		
a. Three or more thread ends not trimmed to a maximum length of 1/4 inch or three or more thread scraps not removed from the liner; any area of any hook fastener tape, one inch or more in length, that contains thread scraps lint or other impurities affecting use		X
b. Any unsightly slub, spot, or stain; any discoloration of the cloth caused by the removal of any shade and size marking ticket; any shade and size marking ticket; excluding the metal fastening or sewn-on kind, not removed; any shade and size marking identification stamping visible on the outerside of the liner; or any shade and size marking adhesive mass adhering to the cloth		X
c. Any metal fastening device or sewn-on shade and size marking ticket attached to the liner	X	

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TABLE II. Classification of defects for visual examination of the assembled liners. (Continued)

Defect	Major	Minor
d. Any marking used for locating the position for attaching any component visible on the top or underside of the liner		X
SLIDE FASTENER		
a. Any surface rough, misaligned, distorted, or contains any pit, nick, sliver, burr, sharp edge, dent, or crack	X	
b. Any surface unclean or contains embedded foreign matter		X
c. Slide fastener not specified type, size, style, pull, or any part of the assembly omitted, bent, or broken 2/	X	
d. Any portion of the liner or the slide fastener tape puckered, twisted, or pleated at the slide fastener tape joining seam, when the slide fastener is open or closed		X
e. Movement of any slider interfered with or scoops do not mesh or interlock	X	
f. Slider not located at the specified position, when the slider fastener is closed	X	
g. Any part not uniform in color or not specified color		X
h. Any thong omitted from the slide fastener, not securely knotted to any pull or less than 2-1/4 inch or more than 2-3/4 inch in length		X
FASTENER TAPES		
a. 10 to 25 percent of the hooks or pile flattened or missing from any tape		X
b. 25 percent or more of the hooks or pile flattened or missing from any tape	X	
c. Any not specified width or length	1/	
d. Any misplaced not serving the intended purpose	X	
e. Any area of any hook or pile tape not one continuous length or width (i.e., length or width achieved by joining together 2 or more pieces		X

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TABLE II. Classification of defects for visual examination of the assembled liners. (Continued)

Defect	Major	Minor
LABEL		
a. Size missing, incorrect, or illegible or label not as specified	X	
b. Other than the size information, identification information incomplete, incorrect, or illegible		X
c. Misplaced, off center of the back by more than 1/2 inch		X
d. Stitching omitted on one or more edges		X
e. Any washing information missing, incorrect, incomplete, or illegible	X	
f. Size of label or marking not as specified		X
COMPONENTS AND ASSEMBLY		
a. Any component not as specified or any defect of component or defect of assembly, not herein classified	1/	
b. Any component, component part, or required operation omitted, or any operation improperly performed, not herein classified	1/	
c. Any component part not cut in accordance with the patterns (see 3.4.1), not herein classified	X	
d. Any dimension not as specified, not herein classified	1/	
SEAMS AND STITCHING		
a. Accuracy of seaming:		
1. Any seam, hem, or attachment of any component twisted, puckered, or pleated; any part of the liner or of any component caught in an unrelated operation or stitching; any seam edge not properly forced out or contains any fold	X	

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TABLE II. Classification of defects for visual examination of the assembled liners. (Continued)

Defect	Major	Minor
2. Ends of stitching that are not caught in other seams or stitching:		
(a) Any not securely backstitched	X	
(b) Any secure but backstitched for a distance less than 1/2 inch or the backstitching is not overlapped on itself		X
3. Thread breaks:		
(a) Any not securely overstitched	X	
(b) Any secure but overstitched for a distance less than 1 inch		X
b. Gage of stitching 3/:		
1. Any stitching irregular, uneven, not gaged as specified, or not uniform		X
c. Open seams 4/:		
1. Any more than 1/8 inch but not more than 1/4 inch		X
2. Any more than 1/4 inch	X	
d. Raw edges 5/:		
1. Outerside or innerside of the liner:		
(a) Any more than 1/4 inch but not more than 1/2 inch		X
(b) Any more than 1/2 inch	X	
e. Seam and stitch type:		
1. Not specified seam or stitch	1/	
2. Any looper thread on the outerside of any liner seam when Stitch Type 401 is used	X	
f. Stitch tension:		
1. Any loose stitching 6/	1/	
2. Any tight stitching resulting in breakage of the stitches when normal pull is applied 7/	X	

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TABLE II. Classification of defects for visual examination of the assembled liners. (Continued)

Defect	Major	Minor
g. Stitches per inch 8/:		
1. Stitches more or less than specified:		
(a) Liner (excluding label):		
(1) One or two		X
(2) More than two	X	
(b) Label:		
(1) One to three		X
(2) More than three	X	
h. Bartack.		
1. Missing, loose, misplaced or not serving intended purpose	X	
2. Length or width not as specified		X

1/ The defect shall be classified and scored as a major defect when it seriously affects the serviceability or appearance, otherwise it shall be classified and scored as a minor defect.

2/ Each slide fastener shall be checked for proper function by opening and closing each slide fastener at least three times along its full length.

3/ The gage of stitching defect shall be scored only when the condition exists for more than 2 inches on any row of stitching.

4/ A seam shall be classified and scored as open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam shall be classified and scored as open when either one or both sides of the seam contain any of the aforementioned for a single stitched seam.

5/ Any edge that should be turned under and is not but is securely caught in the stitching shall be classified and scored as a raw edge. Any raw edge that is not securely caught in the stitching shall be classified and scored as an open seam.

6/ A seam shall be classified and scored as loose when the top and bobbin threads are on the same surface.

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7/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident and is not caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching by pulling the material taut to straighten out the seam. Puckering due to the gathering of the material shall be classified and scored as an accuracy of seaming defect.

8/ The stitches per inch defect shall be scored only when the condition exists for more than 3 inches on any row of stitching.

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TABLE III. List of defects for packaging

Item	Defect
Exterior and interior markings	Missing, incorrect, incomplete, or illegible; of improper size, location, sequence, or method of application; size shown on any interior or exterior container not the same.
Materials	Any noncomforming component; any component or component part missing, damaged or otherwise defective.
Workmanship	Inadequate application of the components, such as incomplete closure of any unit container, container flap, or loose strapping; bulging or distortion of any container; and polyethylene bag damaged, any open or non-continuous heat sealed seam or no 1/4 inch hold in one corner.
Exterior and interior weight or content	Gross or net weight exceeds the requirement; number per container more or less than specified; more than one size liners in the same exterior container; any liner not folded as specified or not individually packaged; any deformed, distorted or bent or any portion of any liner caught in any heat sealed seam.

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